Analysis of a Voltage Controlled Frequency Foldback Technique that Improves Short Circuit Protection for Buck Derived Converters

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Abstract — The output VI characteristics of most high frequency buck derived pulse width modulated (PWM) converters exhibit a current tail during fault conditions. A current tail is actually an increase in the average output current as the output voltage decreases toward zero. Ideally the controller's peak current limit function should prevent the average output current from exceeding a specified limit. In reality, the tail is due to the turn-off propagation delay inherent in the controller and in the power switch. A frequency foldback scheme is proposed to reduce the effects of the propagation delay during an output overload or fault condition. Frequency foldback refers to decreasing the PWM's oscillator frequency during short circuit or overload conditions and allows the duty cycle to decrease below the value normally limited by the total propagation delay.

I. INTRODUCTION

Peak current mode is one type of control method used in Pulsed Width Modulated (PWM) converters. A simplified block diagram of a buck converter using this control method is shown in Fig. 1. The input voltage generator, $v_g(t)$, is typically a constant dc value but may vary within a predetermined minimum to maximum range for a given design. Also shown is the dc resistance of the inductor, R_L, and the equivalent series resistance of the filter capacitor, R_C. The total load resistance is designated by R. Note that the PWM latch is reset dominate indicating that zero duty cycle is possible if the error amplifier output is less than V_{bias}. This topology may also be referred to as a current programmed pulse width converter since the compensated error amplifier output, $v_c(t)$, provides a controlled voltage level which is compared to the voltage analog of the inductor current. The current sense gain, R_i, includes any gain factors such as current transformer turns ratio and resistor divider networks [1]. During normal operation the clock sets the latch at the beginning of a cycle and the MOSFET (will also be referred to as a switch) is turned on. When the voltage analog of the inductor current (plus a dc bias) reaches the error amplifier output, $v_c(t)$, the switch is turned off. It is clear that a current limit is required to prevent failure of the power stage components. A current limit comparator overrides the error amplifier command during large signal transients by resetting the PWM latch through the OR gate. An example of large signal behavior would be an overload at the converter output. For simplicity the soft-start feature, which is often necessary in a practical converter, is not shown in Fig. 1. Note also that Leading Edge Blanking (LEB) is not employed in the current sense path. This feature is often used to blank-out the leading edge spike when the switch turns on, which then permits a smaller current sense filter. A smaller RC current sense filter reduces the phase lag and improves the small signal characteristics of the converter. However, with LEB at higher frequencies, the blanking time variation is more noticeable and may limit the minimum attainable duty cycle for a given design. For this reason LEB is not often used at higher frequencies.

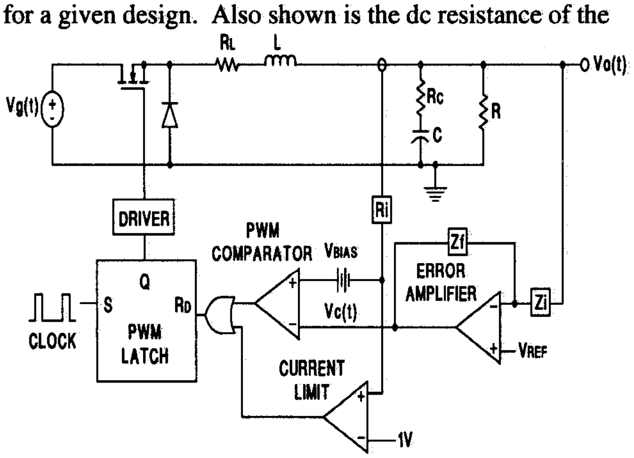

Fig. 1: Model of a Typical Peak Current Mode Controller Shown with a Buck Converter

In practical converters there exists a propagation delay between sensing when the inductor current reaches its limit and when the switch is completely turned off. The total delay is comprised of the current sense to output delay within the controller and the turn-off time of the MOSFET. Typical controllers can have as much as 80ns of delay between the current sense and output pins. A MOSFET (die size 4) can have a total gate charge as high as 70nC with the switching time being dependent on the

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drain-to-source voltage and the gate drive current. If one assumes a constant 1.0A drive the MOSFET turn-off time may be approximated as 70ns. For this case the total system propagation delay is about 150ns. It turns out that this delay sets a limit on the minimum achievable duty cycle. Under normal operation the range of control usually exceeds this minimum duty cycle and it doesn't present a problem. However, during output overload conditions when the output impedance (R) approaches zero ohms the voltage drops out of regulation. This drives the error amplifier into saturation at its high rail (typically 4.7V) as the voltage loop commands for a larger duty cycle. The current limit overrides the error amplifier and terminates the cycle, but only after the total propagation delay will the MOSFET be off. This system behavior causes the peak inductor current to exceed the current limit value which results in overcurrents beyond the intended maximum level. The average output current also increases as the output voltage decreases in almost a runaway situation [2]; this increase in current as the output voltage falls will be called a current tail.

Power converters are being designed at higher operating frequencies in order to reduce the size of the magnetic components. Commercial, high frequency, hard switching converters operating in the range from 400 to 600 kHz are available. The reason can be seen from the transformer power capability equation,

$$P = 0.707 \cdot f \cdot J \cdot W \cdot a \cdot B \times 10^{-8}, \qquad (1)$$

rewritten in the form of the Wa product,

$$W \cdot a = \frac{P \cdot 10^{8}}{0.707 \cdot f \cdot J \cdot B}. \qquad (2)$$

Where P is the power in VA, f is the frequency in Hz, J is the current density in A/cm^2, W is the core window area in cm^2, a is the core cross-sectional area in cm^2, and B is the flux density in Gauss. Smaller Wa product means smaller cores. For constant output power and current density as the frequency is increased the Wa product is reduced. Of course higher core losses occur at higher frequencies which would force a smaller flux density swing. Certainly an upper frequency limit exists due to core losses but less lossy core materials would permit even higher frequency operation.

It is at these higher frequencies that the propagation delay becomes a more significant portion of the operating period. With a 500kHz oscillator a 150ns time delay is 7.5% of the period. With a momentary dead short applied at the output, for example, the excessive currents could stress the semiconductors of the power stage and reduce the converter's life. For these reasons the effect of the current tail should be reduced. One can reduce the propagation delays, and this can help, but it is impossible to eliminate the delay. Reduced delays can require higher performance components that would increase system cost. It is important to remember that the minimum duty cycle limit, caused by the propagation delay, is the original problem which needs to be solved. One possible method to reduce the effects of the minimum achievable duty cycle problem will be described shortly.

Decreasing the operating frequency during overload or fault conditions can reduce the current tail. To understand how this helps it is essential to realize that the propagation delay really sets the minimum achievable duty cycle. The duty cycle is the ratio of switch on-time to operating period. Since the worst-case minimum on-time is the total propagation delay then only the period can be increased in order to decrease the duty cycle. When increasing the period the frequency is decreased and this is referred to as frequency foldback. Frequency foldback only occurs during an overload or fault condition, otherwise the converter operates at a fixed frequency. It follows then that fixed frequency converter and loop design still applies. In summary, an effective way to achieve lower duty cycles and lower average output currents during an overload or short-circuit fault is to increase the period by decreasing the frequency. This frequency foldback scheme will be explained further in the next section.

II. FREQUENCY FOLDBACK TECHNIQUE

The proper configuration of the oscillator section of a PWM controller is critical in achieving desirable frequency foldback characteristics. As the output load increases during an overload the average voltage decreases in a continuous fashion. If the frequency reduction is not continuous an interaction between the power stage and controller is possible [3], potentially leading to latchup as shown in Fig. 2. The discontinuity in frequency can cause a discontinuity, or notch, in the converter's average output current. The load depicted in Fig. 2 is nonlinear and is

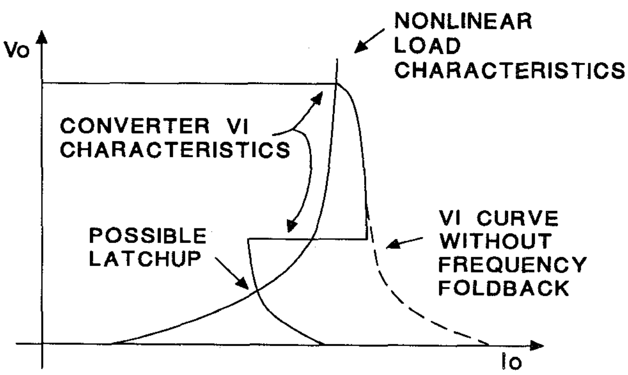

Fig 2: Average Output Voltage vs. Average Output Current for Non-Continuous Frequency Foldback

shown intersecting this notch to emphasize a worst-case situation. Latchup occurs when the load current exceeds the converter's output capability (as indicated in the figure). This happens as the converter increases the output voltage from zero, but at some point the load demand exceeds the converter's ability. The converter can not increase the voltage further and latchup occurs. In addition to latching during an output overload it may also do so during startup conditions. It is desirable to avoid the notching in the converter's output VI characteristic which helps guarantee improved converter operation. Therefore, the rate of change of frequency during frequency foldback should be continuous. Since the current tail increases as the output voltage falls it also makes sense to use the decreasing output voltage to reduce the operating frequency. Fig. 3 shows the frequency foldback characteristic used in this paper. Intuitively, it is best to have a larger df/dV_o (i.e., change of operating frequency per change of average output voltage) at lower values of average output voltage because this is the region where the current tail is most prevalent (as shown by the dotted curve in Fig. 2 for a converter without frequency foldback). The desired increase in df/dVo as Vo decreases, used to reduce the current tail, is achieved within the oscillator section of the controller.

Another desirable feature for peak current mode controllers, under normal steady-state conditions, is to have a linear oscillator ramp that can be used as the source for slope compensation. Slope compensation is only necessary in converters reaching 50% duty cycle or more to ensure stability (assuming trailing-edge modulation [1]). Therefore, this oscillator design will include a linear ramp which can be summed into the current sense circuit to provide for slope compensation.

The circuit topology of Fig. 4 creates a linear increasing ramp and a linear decreasing ramp that forms one cycle of the oscillator period referred to as T_S. The

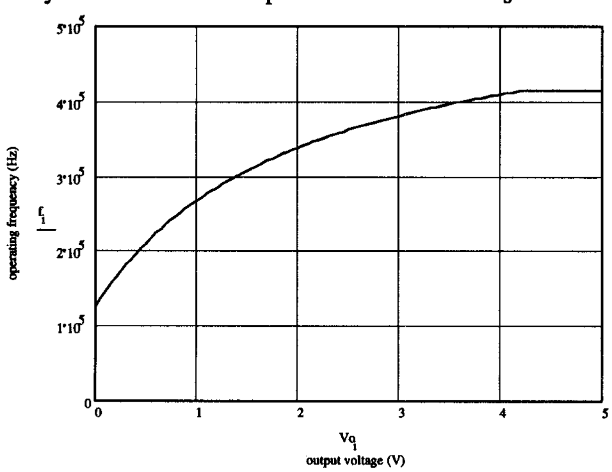

Fig 3: Desirable Characteristics for Frequency Foldback Showing Increasing df/dVo as Vo Approaches Zero

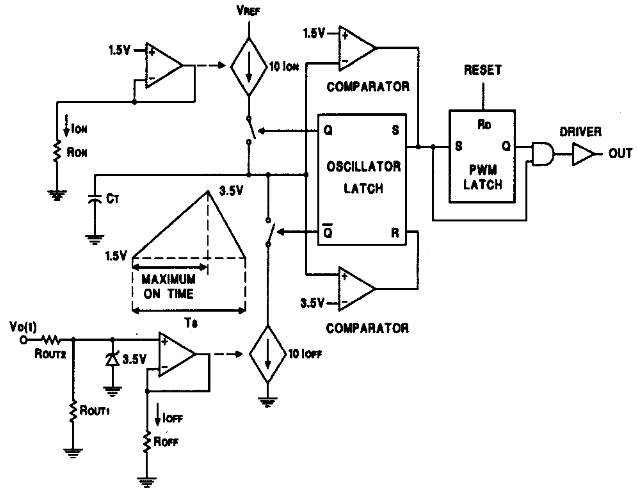

Fig. 4: Oscillator Design Achieves Frequency Foldback

oscillator waveform is designed to have a fixed 3.5V peak and a 1.5V valley. The dependent current source ($10 \bullet I_{on}$) that charges the oscillator timing capacitor, C_t, is fixed by the designer when R_{on} is selected. Since the charge current is constant for a given design, the time to ramp from 1.5V to 3.5V is also fixed. Once C_t is charged to 3.5V, the discharge dependent current source ($10 \bullet I_{off}$) decreases the oscillator timing capacitor's voltage to 1.5V. Note that the discharge current is proportional to the output voltage divided down by programming resistors R_{out1} and R_{out2}. (A third resistor, not shown, connects to the common node formed by R_{out1} and R_{out2} and then to a 5V reference which avoids extremely low frequency operation when $v_o(t)$ is zero volts.) These resistors allow the power supply designer to configure the shape of the frequency foldback characteristic as described above and shown in Fig. 3. In essence the discharge time can increase as the output voltage decreases. The onset of frequency foldback can be forced to occur below the nominal output voltage $v_o(t)$, e.g., 4.2V for a 5.0V output. The advantage here is to keep constant frequency operation within a specified range of the nominal dc output voltage. A comparator is used to set the oscillator latch in Fig. 4 and this same signal (referred to as the clock signal in Fig. 1) is used to set the PWM latch.

The oscillator can be exploited for other hidden capabilities. The ratio of the positive ramping time to total period can be used as a maximum duty cycle clamp and is most beneficial during non-frequency foldback conditions. This is easily done by a logical AND of the Q output of the PWM latch and the comparator output as shown on the right side of Fig. 4. The combined signal then drives the output power stage switch. The duty cycle clamp feature doesn't interfere during overload conditions when the oscillator frequency has decreased, since the peak current limit is in control and terminates each cycle.

A computer program was written to generate the non-linear VI curves of the frequency foldback controller. The advantage of using such a program is that it facilitates fast simulations [4],[5] between circuits with different propagation delays and frequency foldback characteristics under overload conditions. Quick transient analysis is required in order to generate the VI curves in a reasonable amount of time. Other general purpose circuit simulators such as PSPICE and Saber may be used, but they do not generate the VI curve directly. A flow chart of the computer program and representative VI characteristics are given in the next section.

III. COMPUTER MODEL AND SIMULATIONS

A VI curve shows the average output voltage on the ordinate vs. the average output current on the abscissa. It actually hides the voltage ripple, but under normal operation this ripple is usually less than 5% of the output voltage. The curve has two general regions, a constant voltage range where the voltage loop has control and a current limit region which occurs under overload or start-up conditions. As explained previously, the average output current actually increases as the output voltage decreases and this behavior is referred to as the current tail. Generating these curves requires a large signal analysis. The converter circuit does not lend itself well to a closed form equation describing the VI characteristic, which is the main reason for using computer simulations. These simulations also enable one to quickly compare and quantify performance under different system parameter values. The VI curve can be generated by sweeping the load resistance from its maximum value to almost zero ohms. For each load impedance value, the steady state average output voltage and average output current is saved for plotting. After trying several different computer programs, the most efficient method used was a dedicated C program. It quickly does a transient analysis for each load resistance value and saves the steady state average voltages and currents. The load resistance is then reduced and another transient analysis is performed. After a hundred or more analyses the VI curve is smoothly drawn.

A more detailed controller block diagram is shown in Fig. 5 which includes a frequency foldback oscillator model, consisting of an adder and divider, limiter, frequency generator, and a clock with adjustable off-time. The 5V reference is summed with the output voltage in order to avoid extremely low frequency operation when the output voltage is zero. This was used in the laboratory circuit to set the minimum oscillator frequency. A limiter follows this stage to clamp the output of the adder to 3.5V. The 3.5V maximum limit can be used to allow for some output voltage droop before

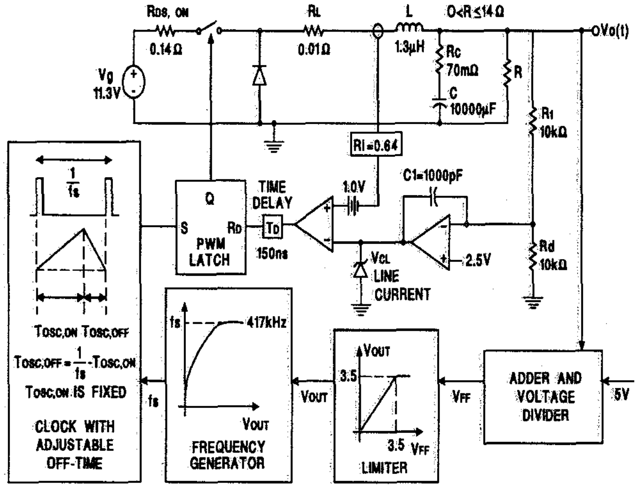

Fig. 5: Detailed Model for Computer Simulation of Frequency Foldback Control

frequency foldback occurs as shown in Fig. 3. A frequency generator is next and it determines the frequency for each cycle based upon V_{out}, the output of the limiter. Within the clock or oscillator block, the positive going ramp time duration ($T_{osc,on}$) is fixed as opposed to the negative going ramp time ($T_{osc,off}$), which increases as the output voltage decreases. Only during frequency foldback does the period of oscillation increase. A time delay is also shown in Fig. 5 which simulates the total propagation delay mentioned earlier.

A flowchart highlighting the computer algorithm is given in Fig. 6. Circuit simulation parameters, such as power circuit inductance, capacitance, and total propagation delay are specified before the program runs. The function starts by setting the output resistance to the maximum value. The switch turn-off time is determined from the closed form circuit equations produced in the first topology mode, that is, with the PWM latch output asserted and the MOSFET turned on. It was assumed that the control circuit could be decoupled from the power circuit since the input impedance of the control circuit is much higher than the load resistance. The characteristic equation for the first and second topological modes are:

$$L \cdot C^2 \cdot \left(R_C + R\right) \cdot s^2 + \left[L \cdot C + C^2 \cdot \left(R_L \cdot R_C + R \cdot R_C + R \cdot R_L\right)\right] \cdot s + C \cdot \left(R_L + R\right) = 0 \qquad (3)$$

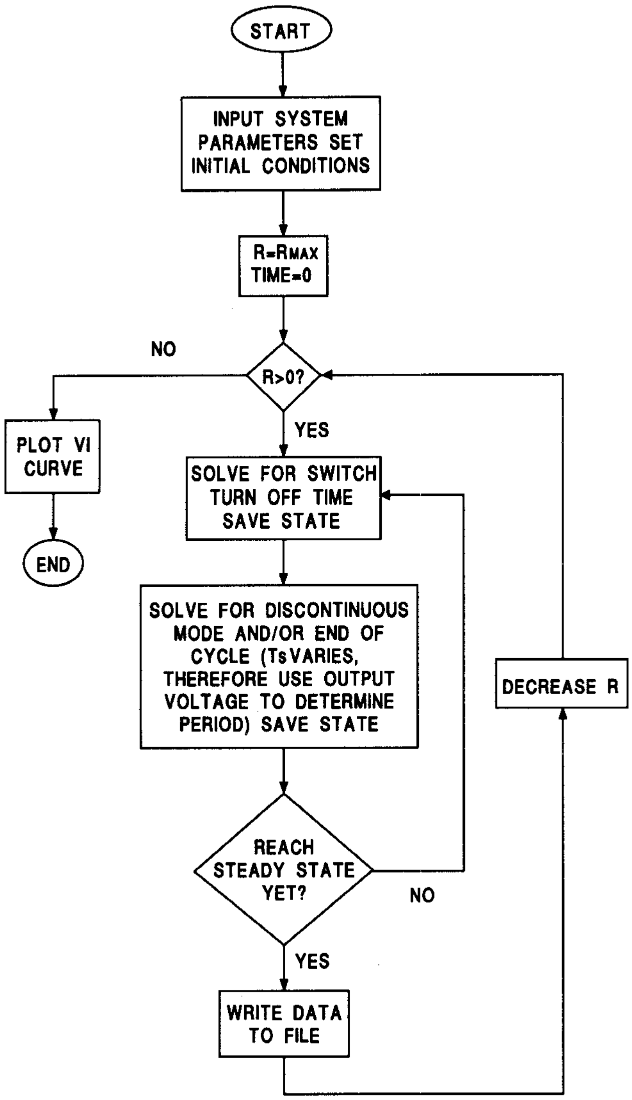

Fig. 6: Simplified Flowchart of Computer Program

which yields the eigenvalues that can now be used to solve for the inductor current. The second topological mode is with the MOSFET off and the diode conducting the current through the inductor. The form of the inductor current equation is, assuming real roots s1 and s2 from (3),

$$i_L(t) = A_1 \cdot e^{s1 \cdot t} + A_2 \cdot e^{s2 \cdot t} + I_f \tag{4}$$

where I_f is the final inductor current (e.g., I_f is given by $v_g/(R_L+R)$ in the first mode) and A_1 and A_2 depend upon the initial conditions at the beginning of the topological mode. Equation 4 can be used for both topological modes 1 and 2 except that A_1, A_2, and I_f will change in each mode. The error amplifier integral compensation produces the following differential equation

$$\frac{dv_c}{dt} = \frac{Vref}{R_d \cdot C_1} - \frac{v_o(t) - Vref}{R_1 \cdot C_1} \tag{5}$$

where Vref is the voltage applied to the noninverting input of the error amplifier and R_d, R_1 and C_1 comprise the compensation network. Equation 4 can now be multiplied by R_i and equated to the solution of equation 5 for $v_c(t)$ giving

$$v_c(t) = R_i \cdot \left[A_1 \cdot e^{s1 \cdot t} + A_2 \cdot e^{s2 \cdot t} + I_f\right]. \tag{6}$$

The unknown in equation 6 is the time when the PWM comparator output gets asserted which is solved for using numerical methods. The first order differential equation (5) may be solved by assuming a small output voltage ripple approximation during the switching cycle. The total propagation delay is then added to the time solved for from equation 6 to give the time instant when the converter actually transitions from mode 1 to mode 2. Once the actual transition time is found, the inductor current and capacitor voltages of the converter are saved. Under the second topology mode, assuming continuous conduction mode, the inductor current is calculated at the end of the period T_S by using equation 4 with appropriate changes in A_1, A_2 and I_f. The value of T_S is based on the present cycle's output voltage as detailed in Fig. 5. If the inductor current just found is greater than zero, continuous conduction mode was valid and the solution for the next cycle can start. Otherwise, discontinuous mode is assumed and the time instant when the inductor current reached zero is solved for by using the Newton-Raphson method. In discontinuous conduction mode (both MOSFET and diode are off and is often referred to as topological mode 3) the final capacitor voltage is given by the equation for a simple RC circuit where the total resistance is R_C+R and the initial capacitor voltage was found at the end of topological mode 2. This cycle-by-cycle transient analysis continues until steady state is reached. Note that it is only necessary to keep the data where the switch transitions take place. A straight line "connecting these dots" is accurate for the time constants used in the power circuit. The final steady state output voltage is saved and the load resistance is decreased by less than 1% and the time transient analysis continues. Upon reaching zero load resistance the analysis is complete and the data is ready for graphing.

The computer simulations were based on the circuit component values shown in Fig. 5. The program was actually expanded to simulate a forward mode converter with a transformer turns ratio of 4 to 1 (buck derived topology) since this was the topology tested in the laboratory. Given 150ns of total propagation delay the results for two VI curves are shown in Fig. 7. For the

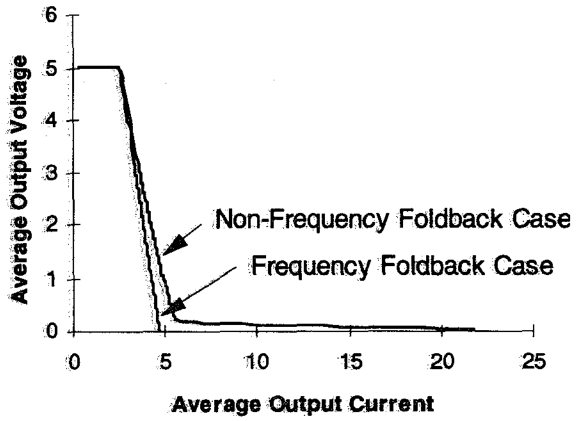

Fig. 7: Frequency and Non-Frequency Foldback Comparison

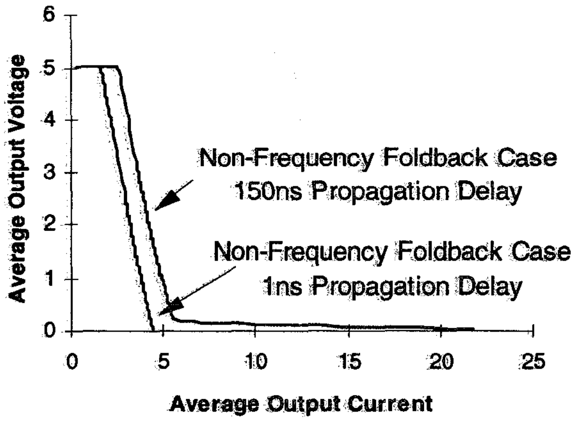

Fig 8: Non-Frequency Foldback with 1.0ns and 150ns Propagation Delays

fixed frequency case, referred to as non-frequency foldback in the figure, the VI curve has a current tail growing as the output voltage decreases. The simulation trace with frequency foldback has no current tail and was produced using a frequency reduction factor of 3.3. The frequency reduction factor is the nominal frequency divided by the minimum frequency. These simulations do not take into account saturation of either the inductor or the transformer, but permit relative comparison between the two methods (with all other things being equal) which highlights the benefits of frequency foldback. Effects of propagation delay can be seen most vividly in Fig. 8. Both VI curves are created with the same circuit parameters and without frequency foldback, i.e., constant frequency even under overloaded conditions. The only difference between the traces is the propagation delay; one with 1.0ns (essentially zero) and the other with 150ns. This graph clearly shows that the propagation delay is the culprit that creates the current tail.

IV. EXPERIMENTAL RESULTS

A 400kHz RCD clamp forward mode converter was designed and tested in the laboratory to verify the concept and the theoretical analysis of this frequency foldback approach. The power circuit values are the same as what is shown in Fig. 5 [6]. The current sense resistor was increased in order to facilitate laboratory current measurements at reasonable levels. It should be pointed out that the higher short circuit currents would have pushed the magnetics deeper into saturation and thus lead to more severe current tails than what was measured. The prototype circuit was configured such that the frequency foldback capability could be disabled by moving a single switch with control power applied. A fan was used to keep the output current resistive shunt stabilized in temperature to increase measurement accuracy.

Fig. 9 shows the data taken with and without frequency foldback. It was collected by first setting the maximum load resistance with frequency foldback enabled, then measuring the average output voltage and current. The switch was then moved to the non-frequency foldback position and the average voltage and current was measured again. The cycle was repeated by first decreasing the load resistance and then measuring the average voltage and current with and without frequency foldback until the minimum resistance was reached. All the data was taken using a prototype of the UCC3884 frequency foldback integrated controller. It had a 0.7V lower limit on the VOUT pin which is used to decrease the frequency [6]. Because of this, the minimum achievable frequency in the laboratory was 267kHz. This limitation will not be in the final controller and a frequency reduction

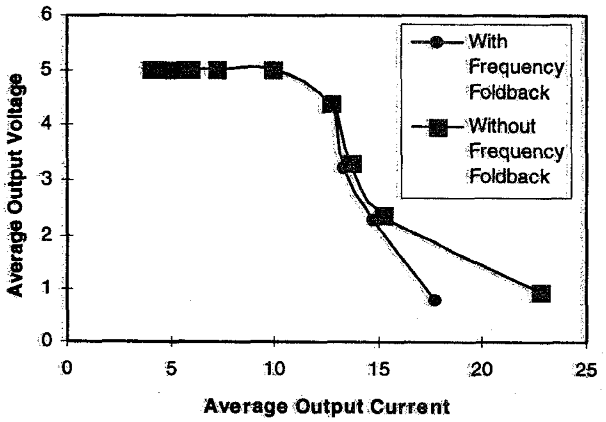

Fig 9: VI Laboratory Data with and without Frequency Foldback

factor of up to about 7 to 1 should be realized. It is reasonable to expect that the small current tail on the frequency foldback curve shown in Fig. 9 would decrease when the VOUT pin limitation is removed. All the frequency foldback computer simulations used a frequency reduction factor of 3.3 to 1. Therefore, as a first order approximation, the nominal 400kHz frequency used in the application circuit would have needed to be reduced to 120kHz, in order to be more effective.

The horizontal scales between Fig. 9 and Fig. 7 do not match due to the different scaling in the current sense network and RC filtering delays in the real circuit. The purpose of the computer model was to compare, without saturation and other second order effects, the relative merits of frequency foldback vs. non-frequency foldback operation. Both the computer model and the laboratory circuit showed that the total propagation delay can make a significant impact on the VI characteristic of buck derived converters and frequency foldback can provide a viable solution.

V. CONCLUSION

High frequency buck derived peak current mode converters can benefit from using the frequency foldback technique during overload and short circuit conditions. The computer simulations and laboratory data confirmed that the current tail can be reduced in a desirable fashion by smoothly decreasing the operating frequency as the output voltage decreases using peak current mode control. Only under fault conditions when the current tail forms and the power stage semiconductors are stressed, does the operating frequency decrease. Therefore the ease of fixed frequency power circuit design and compensation are maintained for normal operation. Minimum frequencies of 1/3 to 1/7 of nominal operating frequency are most effective in reducing the current tail. A new PWM controller from Unitrode, the UCC3884, implements this frequency foldback technique. It also features a maximum duty cycle clamp and a volt-second clamp which helps to prevent the transformer, in isolated converters, from saturating during load and line transients.

REFERENCES

[1] Raymond Ridley, "A New Small Signal Model for Current Mode Control", Ph.D. Dissertation, Virginia Polytechnic Institute and State University, November 1990.

[2] Richard Redl and Nathan O. Sokal, "Overload-Protection Methods for Switching-Mode dc/dc Converters: Classification, Analysis, and Improvements", PESC, 1987, pp. 107-118.

[3] Brian P. Erisman, Fernando Martin, and Richard Redl, "A Novel Primary-Side Controller and an Error Signal Isolator in Low Cost Five-Pin Packages", Toko America, Inc., Revised from APEC '93 Conference Proceedings, with permission, pp. 367-379, 1993, IEEE.

[4] Henry Shu-hung Chung and Adrian Ioinovici, "Fast Computer-Aided Simulation of Switching Power Regulators Based on Progressive Analysis of the Switches' State", IEEE Transactions on Power Electronics, Volume 9, Number 2, March 1994, pp. 206-212.

[5] Ronald C. Wong, Harry A. Owen, Jr. and Thomas G. Wilson, "A Fast Algorithm for the Time-Domain Simulation of Switched-Mode Piecewise-Linear Systems", IEEE Power Electronics Specialists Conference, June 1984.

[6] Philip Cooke, "The UCC3884 Frequency Foldback Pulse Width Modulator", Unitrode Application Note U-164.

[7] Richard Tymerski and Duwang Li, "State-Space Models for Current Programmed Pulsewidth-Modulated Converters", IEEE Transactions on Power Electronics, Vol. 8, Number 3, July 1993, pp. 271-278.

[8] Keyue Smedley and Slobodan Cuk, "Switching Flow-Graph Nonlinear Modeling Technique", IEEE Transactions on Power Electronics, Volume 9, Number 4, 1994, pp. 405-413.

[9] Leu, Hua, Lee and Zhou, "Analysis and Design of RCD Clamp Forward Converter", VPEC, Virginia Tech and Delta Power Electronics Lab, Blacksburg, VA.

[10] Doug Haynes, "Dealing with dc/dc Converter Reliability", PCIM, April 1989, pp. 67-72.

[11] S. Clemente, B. Pelly and R. Ruttonsha, "A Universal 100 kHz Power Supply Using a Single HEXFET", International Rectifier Application Note 939A, HEXFET Power MOSFET Designer's Manual, 1993, pp. 15-26.